



Parktown Boys' **High School**

Geography Department

Grade 12 June Exam

MEMO

SECTION A: CLIMATE AND WEATHER, GEOMORPHOLOGY, AND SETTLEMENTS

QUESTION 1: Climate and Weather

1.1. Various possible answers are provided for each question. Write down **only the letter** of the correct answer next to the corresponding number.

1.1.1. Air that flows down a mountain and along the valley floors at night is known as a(an) ...

- A. Upslope wind.
- B. Anabatic wind.
- C. Downslope wind.
- D. Katabatic wind. (1)

1.1.2. Which direction should one face their house if they wanted to benefit from more heat in the South African winter?

- A. North
- B. West
- C. South
- D. East (1)

1.1.3. A cold front occlusion happens when....

- A. The cold air is left on the ground.
- B. Air ahead of the cold front is slightly warmer than the air behind the cold front.
- C. Air ahead of the cold front is colder than the air behind the cold front.
- D. The air ahead of the warm front is incredibly hot. (1)

1.1.4. Which of the following is NOT associated with the formation of a tropical cyclone?

- A. Coriolis force
- B. Mixing of the westerlies and polar easterlies
- C. Latent heat
- D. Hot moist air (1)

1.1.5. Which of the following factors would hinder the development of an inversion layer?

- A. Low temperatures
- B. Clear skies
- C. Cloudy skies
- D. Still conditions

(1)

1.2. Match the definition/description in **Column B** with each term in **Column A**.

Write the **letter only** of the correct answer next to the corresponding number.

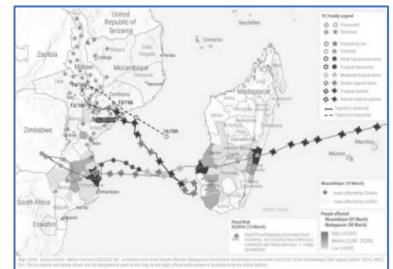
E.g. 2.9. M

Column A	Column B
1.2.1. Moisture front	A. An area on earth where no tropical cyclones form.
1.2.2. Equator	B. The average atmospheric conditions of a small area such as a valley.
1.2.3. Isobars	C. Warm, moist north-easterly winds meet cool south-westerly winds here
1.2.4. Inversion layer	D. A layer of air where the temperature increases with height.
1.2.5. Microclimate	E. Lines that measure the average temperature of an area
	F. An area that does not receive any sunlight
	G. Lines measuring the hectopascals or millibars of an area
	H. The average atmospheric conditions of a large area such as Johannesburg.

(5 x 1) (5)

MCQ 1		Match	
	D	1	C
2	A	2	A
3	B	3	G
4	B	4	D
5	C	5	B

1.3	Refer to the article below on Tropical Cyclone Freddy.		
	Record-breaking Tropical Cyclone Freddy formed on Feb. 3, 2023, northwest of Australia before traveling across the entire Indian Ocean to become the first Category 5 storm of this Southwest Indian Ocean cyclone season. According to the National Oceanic and Atmospheric Administration, no tropical cyclones have taken such a path across the Indian Ocean in the past two decades. On March 7, Freddy became the longest-lived tropical cyclone ever recorded and is now officially Earth's most energetic storm ever observed. Cyclone Freddy made landfall on Madagascar's eastern coast, near Mananjary, on Feb. 21. Freddy then moved across the Mozambique Channel and made landfall in Mozambique's Inhambane province on Feb. 24. Cyclone Freddy made its third landfall in total and its second landfall in Mozambique on March 11 in Zambézia province, with maximum winds of almost 92 miles per hour (148 kilometers per hour). Freddy moved over land as a tropical depression, with a localized center close to the border between Mozambique and the southern tip of Malawi, a landlocked country. Although the system has dissipated, it still generated intense rainfall in the interior of Mozambique and southern Malawi. Source: United Nations Office for the Coordination of Humanitarian Affairs (UNOCHA)		
	1.3.1	What is another term for a tropical cyclone?	(1 x 1) (1)
		Hurricane, Typhoon or Willie Willie ✓✓	
	1.3.2	What number cyclone was tropical cyclone Freddy in this cyclone season?	(1 x 2) (2)
		6 ✓✓	



	1.3.3	“Freddy moved over land as a tropical depression...”. What is a tropical depression? (1 x 2)	(2)
		[Concept] It is a large thunderstorm/heavy storm ✓✓	
	1.3.4	By referencing the path that Freddy followed, explain what was highly unusual about this tropical cyclone. (2 x 2)	(4)
		[Accept any of the following] Freddy made landfall 3 times. / Freddy travelled over Madagascar, then hit Mozam, then returned to the ocean, then hit Mozam again ✓✓ Tropical cyclones usually die after making landfall, Freddy did it 3 times. ✓✓ Freddy re-entered the ocean and made landfall again after that ✓✓ Freddy endured an immense amount of friction and carried on ✓✓	
	1.3.5	List TWO environmental impacts that tropical cyclones can have on a country. (2 x 1)	(2)
		MUST BE ENVIRONMENTAL ✓✓ [Accept any logical answer] Flooding destroys natural habitat Ecosystems are destroyed Coral reefs are destroyed Wild animals killed [DO NOT ACCEPT AGRICULTURE/CROPS/LIVESTOCK]	
	1.3.6	Describe TWO precautions that Mozambique could implement to mitigate the impact of future cyclones. (2 x 2)	(4)
		[Accept logical answers] Early warning Educate the people Prepare evacuation places Reinforce structures Improve drainage etc. ✓✓✓✓	[15]

1.4	Refer to FIGURE 1.4 below showing an urban heat island over a South African city in the peak of summer.		
			
1.4.1	What is an <i>urban heat island</i> ?	(1 x 2)	(2)
	An Urban heat island refers to a situation where the air temperature over a city is warmer than the air in the surrounding suburban and rural areas. ✓✓		
1.4.2	Which area will be warmer at night, the urban or rural areas?	(1 x 1)	(1)
	Urban ✓		
1.4.3	“During summer urban areas receive more convectional rain than rural areas”. Defend this statement with an argument explaining why it is correct.	(2 x 2)	(4)
	[Accept logical arguments] Urban areas generate more heat ✓✓ More cars, more industry/pollution, more energy generated in urban areas ✓✓ More heat leads to an increase in convection currents, which results in more condensation and cloud formation over urban areas ✓✓ Convectional rain is formed by rising air resulting from excess heat, and there is more heat in urban areas ✓✓		

	1.4.4	Write a paragraph of approximately EIGHT lines discussing TWO economic consequences of urban heat islands and TWO solutions we can implement to reduce its effect. (4 x 2)	(8)
		[Accept reasonable answers] ECONOMIC IMPACTS – Must link to the economy or money Increased cost of aircons/cooling Increased precipitation could damage buildings/infrastructure resulting in money being spent to maintain buildings Increased wear and tear on buildings is costly ✓✓✓✓ SOLUTIONS Make surfaces more reflective, more colourful and lighter in colour (dark surfaces hold more heat) We are also Greening the cities Promote greenbelts Legislation to reduce pollution Rooftop gardens Encourage public transport ✓✓✓✓	
			[40]

QUESTION 2: GEOMORPHOLOGY

2.1. Select the correct answer found in the brackets. Write only the **correct question number** and **answer**. i.e. 2.1.8) Rainfall.

- 2.1.1. Ridges of high ground between individual channels of the stream network are called (interfluves/confluence points)
- 2.1.2. Rainwater infiltrates and flows through the soil as (throughflow/permeable flow).
- 2.1.3. A drainage pattern that develops on a dome is a (trellis/radial) pattern.
- 2.1.4. The transverse profile is also known as the (longitudinal/cross-profile) of a river.
- 2.1.5. (Isostatic uplift/Abstraction) is the process whereby the watershed becomes lower and its position shifts.

(5 x 1) (5)

1	Interfluves
2	Throughflow
3	Radial
4	Cross-profile
5	Abstraction

2.2.1. State whether the following statements are **TRUE** or **FALSE**. Write only TRUE or FALSE on your answer booklet next to the corresponding number.

2.2.1. Episodic rivers are never supplied with underground water.

2.2.2. The correct formula for drainage density is:

$$\frac{\text{Total area of the drainage basin (km}^2\text{)}}{\text{Total length of the streams in the drainage basin (km)}}$$

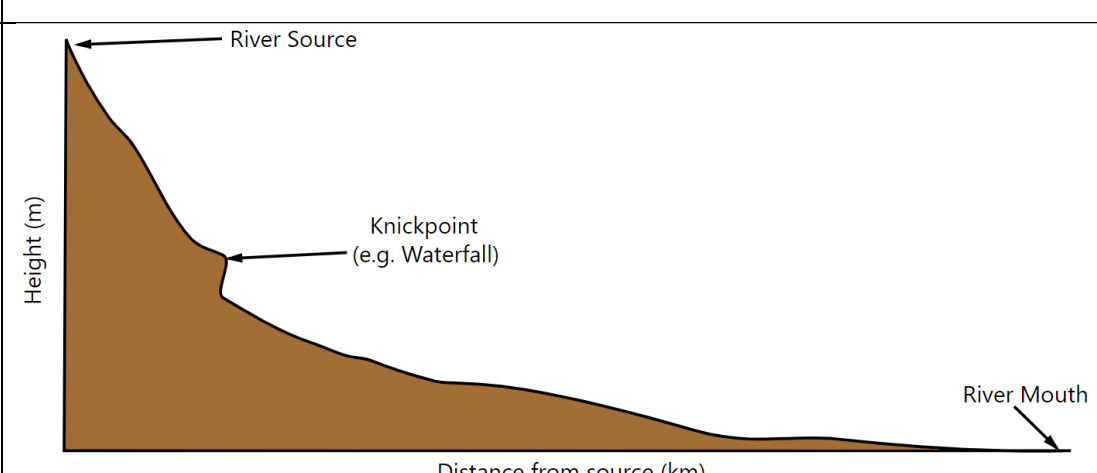
2.2.3. A 4th order stream has a greater volume than a 1st order stream.

2.2.4. Rapids are a feature of the upper course of a river.

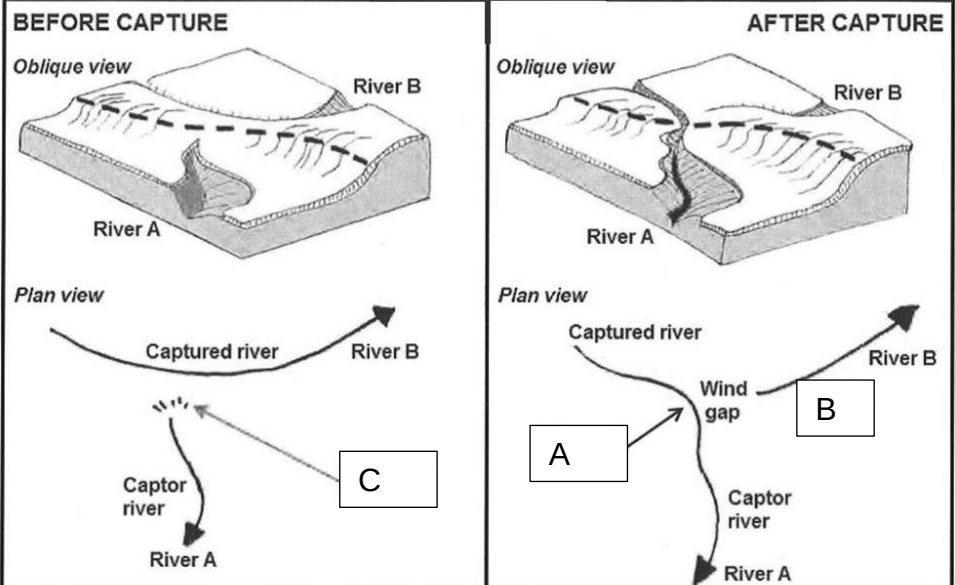
2.2.5. A graded river is convex in shape due to a steep gradient in the upper course and a gentle gradient in the lower course.

(5 x 1) (5)

1	True
2	True
3	True
4	True
5	False

2.3	Refer to FIGURE 2.3. showing a river profile		
			
2.3.1	Define the term <i>longitudinal profile</i> .	(1 x 2)	(2)
	[Concept] A side-view of a river's path showing its flow from its source to the mouth ✓✓		

2.3.2	Redraw FIGURE 2.3 above showing the upper, middle and lower courses respectively. NOTE: You will be marked on approximate placement of these labels, they do not have to be exact. (3 x 1)	(3)
	Upper ✓ Middle ✓ Lower ✓	
2.3.3	Would the above profile be an example of an <i>overgraded</i> or <i>undergraded</i> river? (1 x 1)	(1)
	Overgraded ✓	
2.3.4	Provide ONE example of a temporary base level from FIGURE 2.3 above. (1 x 1)	(1)
	Knickpoint ✓	
2.3.5	Differentiate between <i>turbulent</i> and <i>laminar</i> flow. (2 x 2)	(4)
2.3.6	Draw a fully labelled cross section of a meandering river. Your diagram should include the following labels: Slowest current Fastest current Slip-off slope Undercutting (4 x 1)	(4)
	Slip-off Slope ✓✓✓✓ Slow flow Fast flow Undercutting	[15]

2.4	Refer to FIGURE 2.4 showing river capture below.		
			
2.4.1	Define the term <i>rejuvenation</i> .	(1 x 2)	(2)
	[Concept] A river gains more energy/water ✓✓		
2.4.2	Describe the process happening at C.	(1 x 2)	(2)
	[Concept] Headward erosion ✓✓		
2.4.3	Write a short paragraph analysing the impact of river capture on both human settlement and the natural wilderness living downstream of River B.	(4 x 2)	(8)
	[Accept logical arguments] HUMANS Less water Food pressure Increase in demand, decrease in supply Potentially less electricity generation capacity Less tourism Less industrial production etc. ✓✓✓✓ WILD Habitat destruction Death of all animals Ecosystems interrupted Wildlife migration Breeding patterns for fish interrupted ✓✓✓✓		

		After capture, River A will now have more energy and be able to erode through more resistant rock. As time wears on this more power river maintains its course, cutting through resistant rock, while the landscape around it changes. Due to its power this river keeps its shape. River A is now at odds with the geology and relief around it.	
	2.4.4	Is the above description that of a <i>superimposed drainage pattern</i> or an <i>antecedent drainage pattern</i> ? (1 x 1)	(1)
		Superimposed ✓	
	2.4.5	Is it possible to identify whether a drainage pattern is superimposed or antecedent without some prior knowledge of the geological history of the area? Justify your answer. (1 + 1)	(2)
		No. ✓ Both of these patterns require knowledge of the ages of the rock and what the area looked like before ✓	[15]
			[40]

QUESTION 3: SETTLEMENT GEOGRAPHY

3.1. Various possible answers are provided for each question. Write down **only the letter** of the correct answer next to the corresponding number.

3.1.1. In rural areas industrialisation and the modernisation of agricultural practices has led to:

- A. Voluntary resettlement
- B. Less output
- C. Rural-urban migration
- D. Basic need philosophy
- E. Formal resettlement (1)

3.1.2. Which of the following factors is NOT a site factor to consider in the development of a settlement

- A. Availability of water
- B. Rich soils
- C. Flat gradients
- D. Central point of trade
- E. A natural harbour (1)

3.1.3. Which is the smallest settlement?

- A. Hamlet
- B. Village
- C. Isolated farmstead
- D. Conurbation
- E. City

(1)

3.1.4. Which of the following would be considered a high-order good?

- A. An imported sports car
- B. A cheap cell phone
- C. Bread
- D. A delicious pizza
- E. A pair of shoes

(1)

3.1.5. Rural economic development is hindered because of:

- A. These areas are often remote
- B. Transport costs are expensive
- C. These areas are seen as poor investments
- D. Many workers suffer from diseases
- E. All of the above

(1)

1	C
2	D
3	C
4	A
5	E

3.2. State whether the following statements are **TRUE** or **FALSE**. Write only TRUE or FALSE next to the corresponding number in your answer book.

3.2.1. Urban areas are multifunctional in nature.

3.2.2. The site of a settlement means a settlement's location in relation to its surrounding features.

3.2.3. Rapid urbanization is most prevalent in more economically developed countries.

3.2.4. Large scale solar farms would best be built in an urban area.

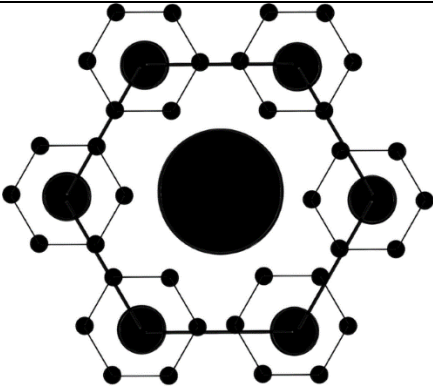
3.2.5. One of the causes of rural-urban migration is a decrease in yield per unit area of agricultural land.

(5x1)(5)

1	TRUE
2	FALSE
3	FALSE
4	FALSE
5	FALSE

3.3	Refer to FIGURE 3.3 below and answer the questions that follow.		
			
	3.3.1	Is FIGURE 3.3 referring to rural or urban settlements? (1 x 1)	(1)
		Rural ✓	
	3.3.2	Explain the humour in the cartoon. (1 x 2)	(2)
		[Accept logical answers] These people do not need technology, they need water ✓✓ The educated scientists have no idea what these people actually need ✓✓ There is a massive lack of communication between the top and the bottom ✓✓	

	3.3.3	The conditions visible in this cartoon have negatively impacted on many site factors that initially contributed to the development of the rural settlement here. List TWO site factors that are being negatively impacted by the conditions seen in the cartoon. (2 x 1)	(2)
		[Any TWO] Soil Microclimate Climate Water ✓✓	
	3.3.4	What is <i>social justice</i> ? (1 x 2)	(2)
		[Concept] Access to basic humans needs ✓✓	
	3.3.5	In a paragraph discuss TWO social injustices that many rural South Africans are facing and TWO possible reasons for this. (4 x 2)	(8)
		[Accept reasonable answers] SOCIAL INJUSTICES Socio-economic and physical issues – Rural is mostly agriculture. These people do not have the finances or the soil, thus lack basics Access to primary healthcare – not enough doctors, thus lack of access to health HIV/AIDS and TB – Again, lack of doctors and medicine Poor rural infrastructure and lack of access to water and electricity – Lack of access to basics Not enough schools also creates a lack ✓✓✓✓ REASONS Not profitable to invest in these areas They are remote, transporting things there is expensive, the roads are bad, there isn't much to gain from travelling there Cities have far more people so the funds go there Corruption and improper spending ✓✓✓✓ [15]	

3.4	Study FIGURE 3.4 below and answer the questions that follow.		
	<u>Central Place Theory</u>		
			
	3.4.1	What is <i>Central Place Theory</i> ? (1 x 2)	(2)
		[Concept] A theory that is used to place the distribution of settlements ✓✓	
	3.4.2	Using the image as a guide, draw a legend/key for this diagram indicating the following labels: Large City Suburb Intermediate Centre (3 x 1)	(3)
		 Large City ✓  Intermediate Centre ✓  Suburb ✓	
	3.4.3	Central Place Theory operates under the premise of low-order and high-order centres. Which type of centre would you classify Johannesburg as? (1 x 1)	(1)
		High-order ✓	
	3.4.4	Provide a reason for your answer to QUESTION 3.4.3. (1 x 2)	(2)
		[Accept any] Large population Large variety of high-order goods Large area Centre for multinational trade ✓✓	

	3.4.5	Differentiate between <i>threshold population</i> and <i>sphere of influence</i> . (2 x 2)	(4)
		[Concepts] Threshold – Number of people needed to maintain a business ✓✓ Sphere – How far a person is willing to travel to get a good/service ✓✓	
	3.4.6	You have inherited a lot of money and wish to open an exclusive diamond jewellery store. An estate agent has presented you two locations namely Sandton and Hillbrow. Which location would you choose to set up your store? (1 x 1)	(1)
		Sandton ✓	
	3.4.7	List TWO centripetal forces that would have influenced your decision to QUESTION 3.4.6 (2 x 1)	(2)
		[Accept any] Functional prestige More business opportunity High-end clients Better foot traffic Better security ✓✓	[15]

TOTAL SECTION A: 120 MARKS

SECTION B: GEOGRAPHICAL SKILLS AND TECHNIQUES (MAPWORK)

RESOURCE MATERIAL

1. An extract from the topographic map 2931 CA VERULAM
2. Orthophoto map 2931 CA 11 VERULAM

GENERAL INFORMATION ON VERULAM



Nestled in the rolling hills of the KwaZulu-Natal coastline, Verulam lies on the banks of the uMdloti River, with the most of the town lying south of the river except the industrial suburbs of Canelands and Barrs Flats. The town is flanked by Waterloo and Ottawa South to the south-east, Phoenix to the south, Mawothi to the south-west and Redcliffe to the west. Neighbouring communities in the greater surrounding area include Mount Edgecombe (10 km) to the south-east, eMdloti (11 km) to the east, uMhlanga (12 km) to the south-east and oThongathi (13km) to the north-east.

Verulam is one of the smaller industrial nodes of the Greater Durban metropolitan area with many industries scattered around the town, mostly concentrated to the north of the town. Notable industries that operate from Verulam include: Shoprite (regional distribution centre), Freshmark- a subsidiary of Shoprite (regional distribution centre), Grafton Everest (furniture), Colgate-Palmolive (oral care), AfriSam (construction materials), Frimax (snacks), Packo (spices) and the Parachute Industries of Southern Africa (sport parachutes).

4.1 MAP SKILLS AND CALCULATIONS

- 4.1.1. Calculate the area covered by the orthophoto map in meters as shown on the topographic map (red block).

Formula: AREA = LENGTH X BREADTH

(4x1) (4)

3.7cm ✓
3.2cm ✓

$$\begin{aligned}
 &1.85\text{km} \times 1.6\text{km} \\
 &1850\text{m} \times 1600\text{m} \checkmark \\
 &= 2\,960\,000\text{m}^2 \checkmark
 \end{aligned}$$

Refer to the orthophoto map and answer the following questions on gradient.

- 4.1.2. Determine the average gradient of the slope for a truck that will transport its cargo (goods) from Spot Height 29 at the industries near 6 (block A5) to Spot Height 164 in Everest heights in block E1 on the **orthophoto** map. Show ALL calculations.

Formula: Average gradient = $\frac{\text{Vertical interval (VI)}}{\text{Horizontal equivalent (HE)}}$

OR

Formula: Difference in Height : Horizontal Distance

$$\begin{aligned}
 \text{Height} &= 164 - 29 = 135\text{m} \checkmark \\
 22.6\text{cm} \checkmark &= 2.26\text{km} = 2260\text{m} \checkmark
 \end{aligned}$$

$$\begin{aligned}
 &135 : 2260 \\
 &1 : 16.74 \checkmark
 \end{aligned}$$

(4x1) (4)

- 4.1.3. Will it be easy or difficult for the truck to transport its cargo over the calculated gradient? (1x2) (2)

Easy $\checkmark\checkmark$

[10]

4.2	MAP INTERPRETATION		
	4.2.1	Refer to block D5 on the topographic map. What direction is the Mdloti river flowing? (1 x 1)	(1)
		West $\checkmark$	
	4.2.2	Provide a reason evident on the map to justify your answer to QUESTION 4.2.1. (1 x 2)	(2)
		[Accept any] Contours Dam wall Angle of tributaries $\checkmark\checkmark$	

	4.2.3	Refer to block E4 . What drainage basin pattern is evident for this river system? (1 x 1)		(1)
		Dendritic ✓		
	4.2.4	Refer to block E4 . Give the stream order of the river exiting block E4. (1 x 2)		(2)
		3rd order ✓✓		
	4.2.5	Refer to the settlement of Hazelmere (A2) and Verulam (South west corner of the map) .		
		(a)	Which of these settlements would be classified as a town and which settlement would be viewed as a village? (2 x 1)	(2)
			Hazelmere – Village ✓ Verulam – Town ✓	
		(b)	Which settlement would have a larger sphere of influence? (1 x 1)	(1)
			Verulam ✓	
		(c)	Which settlement would have the correct threshold population to open a Law Firm? (1 x 1)	(1)
			Verulam ✓	
	4.2.6	K indicates an area of Verulam that has expanded rapidly in the last few years. Discuss ONE issue these residents may face related to rapid urbanisation. (1 x 2)		(2)
		[Accept reasonable answers] Strain on infrastructure Overcrowding Lack of service delivery Social injustice ✓✓		

[12]

4.3	GEOGRAPHICAL INFORMATION SYSTEMS (GIS)			
	4.3.1	(a)	Define the term <i>Buffering</i> . (1 x 2)	(2)
			[Concept] A line used to demarcate an area around a spatial feature ✓✓	

		(b)	Give evidence to show that buffering was implemented in blocks B5 and C5 around the airport. (1 x 1)	(1)
			[Accept any] No development near airport No veg near airport ✓	
	4.4.3		The Orthophoto map 2931 CA 11 VERULAM is secondary data.	
		(a)	What is <i>secondary data</i> ? (1 x 2)	(2)
			[Concept] Data not gathered by directly being there ✓✓ Data that comes from a secondary source ✓✓	
		(b)	Give one advantage of using secondary data. (1 x 2)	(2)
			[Accept any] Easy to access Can get a lot of it No physical danger No field work needed etc. ✓✓	
		(c)	Is the orthophoto map an example of raster or vector data? (1 x 1)	(1)
			Raster ✓	[8]

TOTAL SECTION B: 30 MARKS

Grand Total: 150 Marks

